



C1 Enzymes and metabolism

C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Enzymes are globular proteins that act as biological catalysts, lowering the activation energy of specific reactions by binding substrates at their active site (via the induced-fit model) without being consumed themselves. Metabolism is the sum of all chemical reactions in an organism, organised into anabolic pathways (building up, requiring energy) and catabolic pathways (breaking down, releasing energy). Enzyme activity is affected by temperature, pH, substrate and enzyme concentration, and by inhibitors: competitive inhibitors resemble the substrate and bind the active site, while non-competitive inhibitors bind elsewhere, altering the enzyme's shape.

Worked example: follow the reasoning

A student measures the initial rate of an enzyme-catalysed reaction at increasing substrate concentrations, keeping enzyme concentration constant. At low substrate concentrations, rate rises steeply and roughly proportionally with substrate concentration. Above a certain substrate concentration, the rate levels off at a maximum, $V_{\max}$, of 8.0 arbitrary units per minute, despite further increases in substrate. Step 1: at low $[S]$, many active sites are unoccupied, so more substrate means more successful collisions and higher rate. Step 2: as $[S]$ increases further, a greater proportion of active sites are occupied at any instant. Step 3: at very high $[S]$, essentially all active sites are continuously occupied (enzyme-saturated), so rate cannot increase further even with excess substrate. Conclusion: $V_{\max}$ (here 8.0 units/min) is reached when enzyme concentration, not substrate concentration, becomes the rate-limiting factor.

Quick check · multiple choice

- A competitive inhibitor reduces enzyme activity by:
- A Binding permanently to an allosteric site and changing the active site shape
 - B Structurally resembling the substrate and binding reversibly to the active site, an effect reduced by raising substrate concentration
 - C Breaking peptide bonds within the enzyme
 - D Increasing the enzyme's $V_{\max}$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The activity of a digestive enzyme was measured at pH values from 2 to 10. [7]
Activity was very low at pH 2, rose to a peak at pH 6, then declined steeply and reached zero by pH 9.
- (a) Identify the optimum pH of this enzyme and describe the overall shape of the graph. [2]
 - (b) Explain, in terms of protein structure, why activity is low at pH 2 and at pH 9–10. [3]
 - (c) Predict what would happen to the enzyme's activity if the pH were returned from 9 to 6 shortly after exposure, and explain your reasoning. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) The optimum pH is approximately 6; activity rises from a low value at pH 2 to a peak at pH 6, then falls steeply to zero by around pH 9 (a roughly bell-shaped curve).
- (b) At both very low (pH 2) and high (pH 9–10) pH, the concentration of H^+ or OH^- ions is far from the enzyme's optimum, disrupting the ionic and hydrogen bonds that maintain the enzyme's tertiary structure; this changes the shape of the active site so it is no longer complementary to the substrate, reducing or abolishing catalytic activity (denaturation).
- (c) If the exposure to pH 9 was brief, activity may partially or fully recover on return to pH 6, because the tertiary structure may not yet be irreversibly disrupted; however, if the enzyme was denatured for longer/more severely, the bonds maintaining its shape would be permanently broken and activity would not be recovered even after returning to the optimum pH.

Quick check answer

B. A competitive inhibitor competes with substrate for the active site; increasing substrate concentration allows substrate to out-compete the inhibitor, restoring rate toward V_{max} .

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.